

Castalia

A four-channel electrochemical sensing SoC with a five-hart RISC-V back end

Maxx Seminario

University of Nebraska–Lincoln

September 21, 2026

Where cancers are caught, and whether anyone looks

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

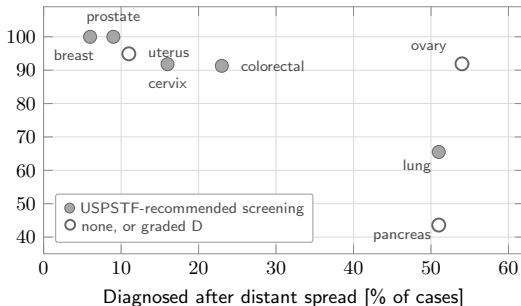
Against Moshkin and the field

From currents to a score

References

Backup

5-year survival if caught localized [%]



[1] SEER 21, 2016–2022; [2, 3] USPSTF

Why ovarian: late, lethal, and unscreened

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

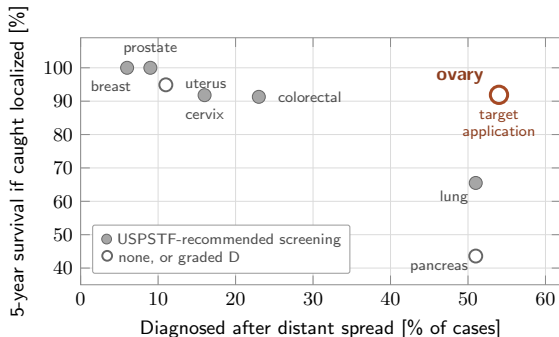
Vision

Against Musklin and the field

From currents to a team

References

Backup



Site	Distant at dx	Deaths/100 cases	Screening (USPSTF)
Ovary	54 %	59	D
Pancreas	51 %	78	D
Lung	51 %	54	B, LDCT
Colorectal	23 %	35	A
Cervix	16 %	31	A
Uterus	11 %	21	none
Prostate	9 %	11	C, PSA
Breast	6 %	13	B

USPSTF: A/B recommended; C offered selectively; D recommended *against*; none: no statement [7].

[1] SEER, 2026 estimates; [2, 3] USPSTF

Most lethal gynecologic cancer [1]: 92 % survivable if caught localized, yet 54 % present after spread, and screening is graded D [2]. UKCTOCS moved stage, not mortality [4].

One serum aliquot, four sites, one score

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

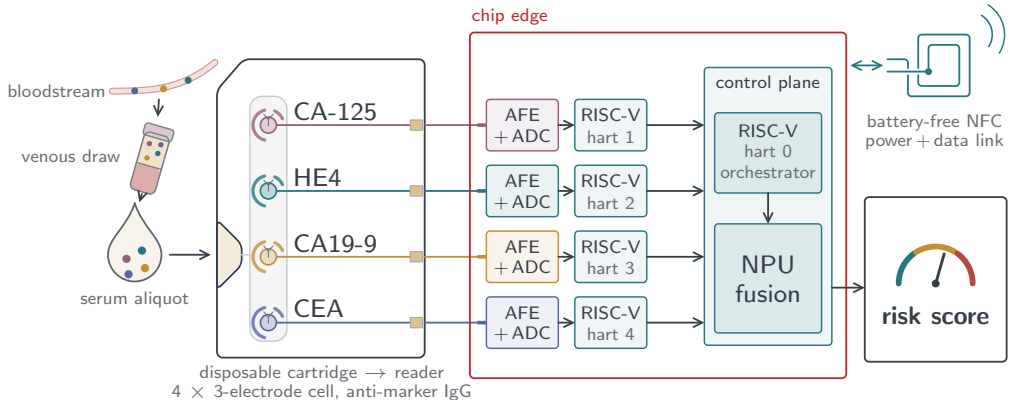
Vision

Against Moshkin and the field

From currents to a score

References

Backup



Symptomatic-patient triage on a disposable four-site cartridge; four markers read at the same instant, fused on chip.

No single marker is a test; four fused are [5]

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

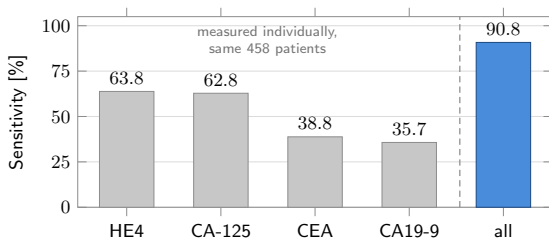
Vision

Against Moshkin and the field

From currents to a score

References

Backup



Marker	Molecule	Cut-off
CA-125	MUC16 mucin epitope [8]	35 U/mL
HE4	WFDC2 glycoprotein [6]	70 pmol/L
CA19-9	sialyl-Lewis ^a antigen	37 U/mL
CEA	CEACAM5 glycoprotein	5 ng/mL

Each marker covers a subtype the others miss. The decision is a learned combination, so the chip is four channels plus the processor that fuses them.

Block diagram

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

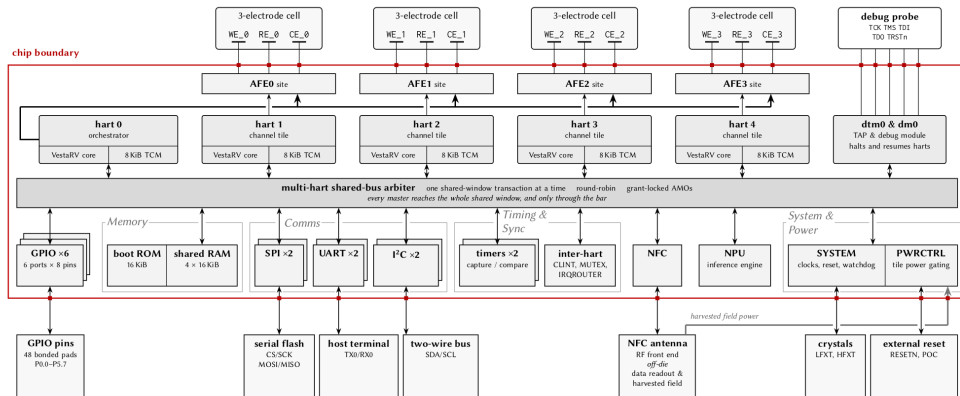
Vision

Against Mwaniki and the field

From currents to vision

References

Backup



Four channel tiles own four AFE sites; every master reaches ROM, RAM, NPU and peripherals only through the arbiter.

Physical implementation

Castalia

Maxx Seminario

Target
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

Against Myshkin
and the field

From currents to
a score

References

Backup

ARTWORK PENDING



**ARTWORK
PENDING**



Grounded-WE two-amplifier potentiostat

Castalia

Maxx Seminario

Target
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

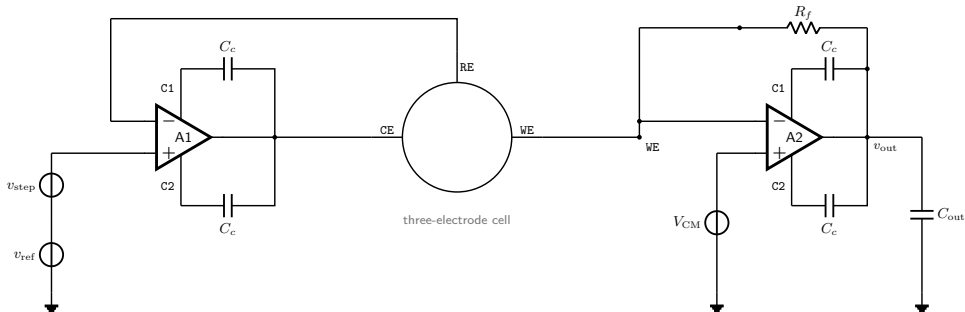
Vision

Against Mvshkin
and the field

From currents to
a trace

References

Backup



A1 drives CE with RE in feedback; A2 holds WE at v_{cm} through R_f . $v_{out} = v_{cm} - I_{we}R_f$, both polarities.

Analog top level

Castalia

Maxx Seminario

Target
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

Against Moshkin
and the field

From currents to
a score

References

Backup

ARTWORK PENDING



Virtuoso export of the analog top level: four channels, shared bias, test bus

The electrode model, in Verilog-A

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

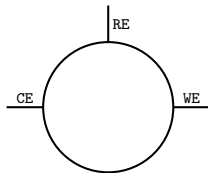
Vision

Against Munkin and the field

From currents to a scan

References

Backup



R_u , R_{ce} , C_{dl} lumped; the faradaic branch Z_f is the Verilog-A element.

i_f faradaic current
 η overpotential, $E - E^{0'}$
 f F/RT
 $C_{O,R}(0, t)$ surface conc. of O, R
 C^* bulk conc.
 δ Nernst layer

Kinetics $i_f = nFAk^0 [C_R(0, t) e^{(1-\alpha)nf\eta} - C_O(0, t) e^{-\alpha nf\eta}]$
 Transport $\partial C / \partial t = D \partial^2 C / \partial x^2$, finite-volume, 28 expanding slabs per species
 Boundaries flux = $\pm i_f / nFA$ at the surface; $C(\delta, t) = C^*$ at the Nernst layer

Parameter	Symbol	Default	Unit	Quantity
E0	$E^{0'}$	0.0	V	formal potential of the couple
nel	n	1	—	electrons transferred
alpha	α	0.5	—	transfer coefficient
k0	k^0	0.1	cm s^{-1}	standard heterogeneous rate constant
Dif	D	5×10^{-6}	$\text{cm}^2 \text{s}^{-1}$	diffusion coefficient, both species
Area	A	0.01	cm^2	electrode area
Cox	C_O^*	1×10^{-6}	mol cm^{-3}	bulk concentration, oxidised species
Cred	C_R^*	0	mol cm^{-3}	bulk concentration, reduced species
Temp	T	298.15	K	interface temperature
expmax	—	40	—	exponent clamp on the kinetic terms
gpar	—	1×10^{-15}	S	DC shunt across the terminals

Equations: Bard–Faulkner [13]. Discretisation: Britz–Strutwolf, Feldberg [17, 18]. Precedent for Verilog-A electrode co-simulation: Jog et al. [19].

Emulating the four markers

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

Against Mushkin and the field

From currents to a score

References

Backup

Marker	Reporter	$E^{0'}$ (V)	Peak (nA)	C_R^* (μ M)	Cut-off
CA-125	methylene blue	-0.15	856	279	35 U/mL [5, 8]
HE4	ferrocene	+0.10	1311	394	70 pmol/L [6]
CA19-9	TMB / HRP [20]	+0.25	1659	458	37 U/mL [5]
CEA	anthraquinone	+0.40	939	303	5 ng/mL [5, 14]

$E^{0'}$ reporter formal potential, i.e. peak position; chosen in the -0.6 to $+0.4$ V window [22, 23], 150–250 mV apart, not measured per reporter

Peak inside 50 nA to 5 μ A on screen-printed electrodes [14, 15, 21]; C_R^* fitted to reach it

k^0 0.1 cm/s, reversible; nothing published on carbon for these markers

Shared $\alpha = 0.5$, $n = 1$, $D = 5 \times 10^{-6}$ cm²/s, $A = 0.01$ cm², 37 °C, $R_f = 560$ k Ω

Representative values, not marker-measured: these slides demonstrate the channel, not an assay.

Channel range, assay range, and where the panel was placed

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

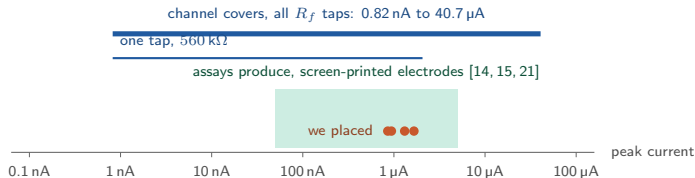
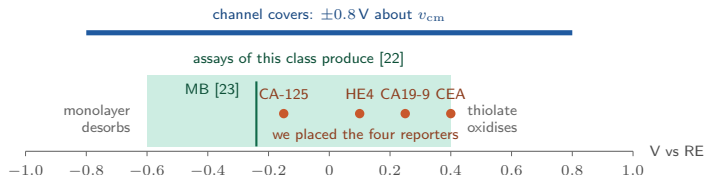
Vision

Against Moshkin and the field

From currents to a score

References

Backup



Potentials: spaced 150–250 mV for one-sweep resolution inside the reporter window, not per-reporter values. Peak heights: inside the published envelope and one tap's full scale, then C_R^* fitted. Silicon is unaffected: the channel covers the assay band with 0.2–0.4 V and a decade of current to spare, so moving the dots changes what the simulation shows, not what the chip can measure.

Exercising the cell: differential-pulse voltammetry

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

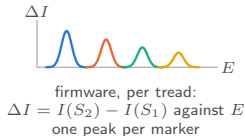
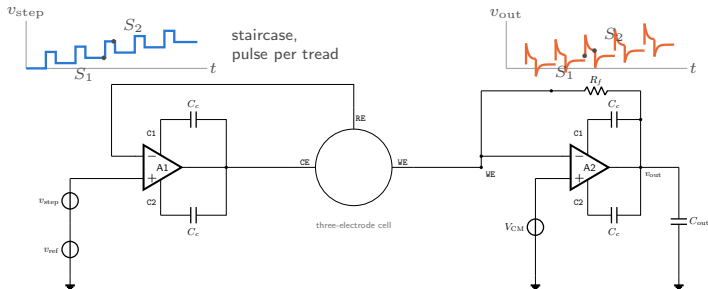
Vision

Against Moshkin and the field

From currents to a score

References

Backup



Excite

DAC: staircase with a pulse on every tread, applied at RE by A1

Sense

$$v_{out} = v_{cm} - i_{WE} R_f$$

Sample

SAR at S_1 (before the pulse) and S_2 (end of the pulse), once charging has decayed

Resolve

firmware: $\Delta I = I(S_2) - I(S_1)$ vs E ; peak position: marker, peak height: concentration

Four markers resolved in one differential-pulse sweep

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

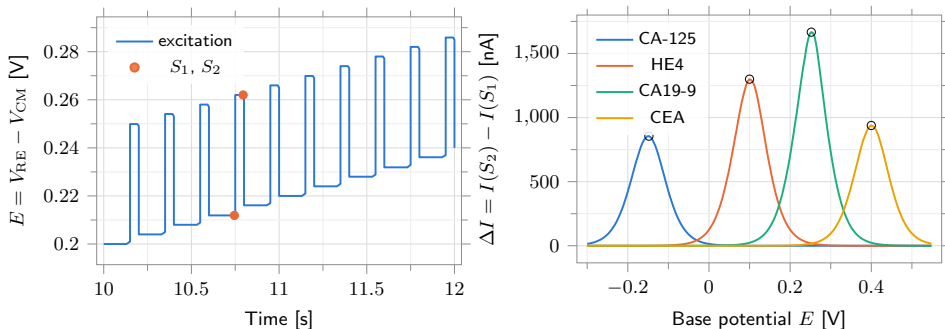
Vision

Against Moshkin and the field

From currents to a score

References

Backup



*schematic sim

Peak position identifies the reporter, peak height carries concentration. Apexes within 2 mV and 1 % of the assay targets; blank below 1 pA.

Every clinical cut-off lands inside the channel window

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

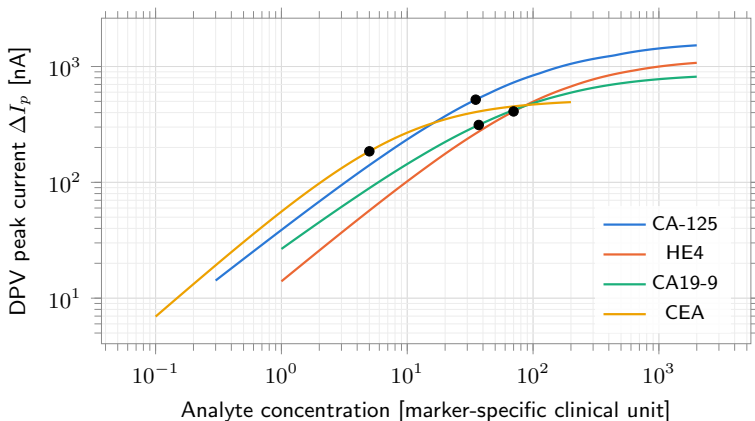
Vision

Against Moshkin and the field

From currents to a score

References

Backup



*schematic sim

CA-125 35 U/mL, HE4 70 pmol/L, CA19-9 37 U/mL, CEA 5 ng/mL: all four on the steep part of the curve at one R_f setting.

Cyclic voltammetry through the assembled channel

Castalia

Maxx Seminario

Target
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

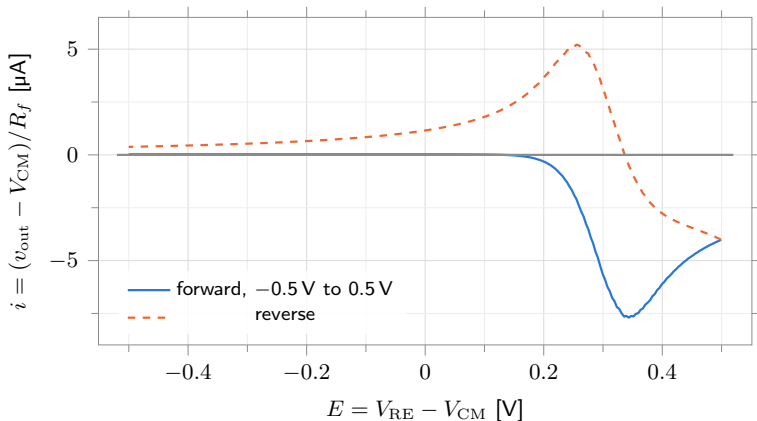
Vision

Against Moshkin
and the field

From currents to
a score

References

Backup



*schematic sim

Staircase potential from the on-chip DAC, current digitised by the on-chip SAR, unclipped over the sweep.

Why impedance: for electrodes that stay in

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

Against Moshkin and the field

From currents to a score

References

Backup

cartridge



single-shot sandwich immunoassay, incubate and read once

wearable

continuous glucose monitors, 10–15 days [27]
wound patch [28]; aptamer sensor, one week in vivo [29]



implant

Eversense implanted glucose monitor, 180–365 days [27]



1 min

1 h

1 day

1 week

1 month

1 year

time the electrode spends in the sample

EIS gives

each site's $R_{ct} \parallel C_{dl}$ arc, trended over the electrode's life, from the same channel
Firmware re-ranges, recalibrates, or retires the site

Drift is systematic: averaging removes noise, not bias [25].

How impedance is measured: the same loop, no extra hardware

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

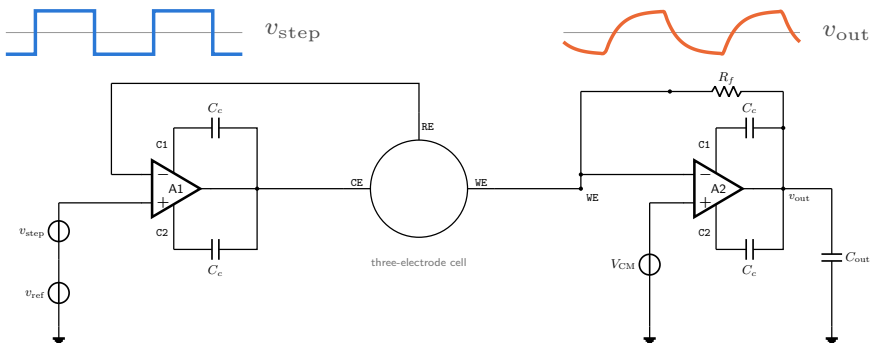
Vision

Against Moshkin and the field

From currents to a score

References

Backup



Excite
Sense
Digitise
Solve

DAC: a small two-code square about the DC setpoint v_{ref}
 $i_{WE} = -v_{out}/R_f$
the SAR records both; firmware I/Q demodulation
 $\hat{Z} = -R_f \hat{V}_P / \hat{V}_{OUT}$, or a ratio to Z_{cal}

Magnitude and phase from codes: I/Q demodulation

Castalia

Maxx Seminario

Target
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

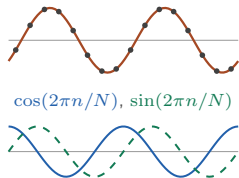
Against Mvshkin
and the field

From currents to
a score

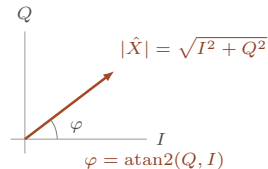
References

Backup

record c_n , N per period, M periods



$$I = \frac{2}{MN} \sum c_n \cos \frac{2\pi n}{N}$$
$$Q = \frac{2}{MN} \sum c_n \sin \frac{2\pi n}{N}$$



Records v_P and v_{out}
Divide $\hat{Z} = -R_f \hat{V}_P / \hat{V}_{OUT}$
Ratio against Z_{cal} : R_f drops out

The four loads behind the EIS plots

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

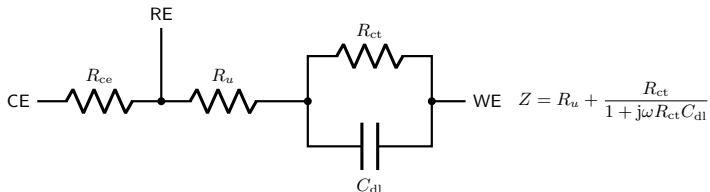
Vision

Against Moshkin and the field

From currents to a trace

References

Backup



Load	R_{ct}	C_{dl}	Gain code	Stands for
resistor	–	–	$N = 0$	an 11 k Ω resistor in place of the cell: no interface, so the trace is the channel's own error
fouled	10 k Ω	10 μ F	$N = 0$	a degraded electrode: adsorbed protein and biofilm open leakage paths and enlarge the interface
typical	100 k Ω	1 μ F	$N = 20$	an electrode in use
fresh	1 M Ω	100 nF	$N = 63$	a new, clean surface

$R_u = R_{ce} = 1 \text{ k}\Omega$ throughout. Fouling: R_{ct} falls and C_{dl} rises through an electrode's life; tracking that drift is what EIS is for here.

Impedance spectroscopy on the same channel

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

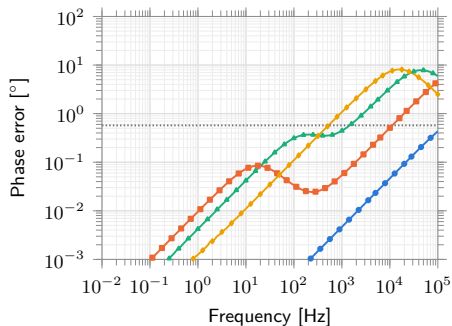
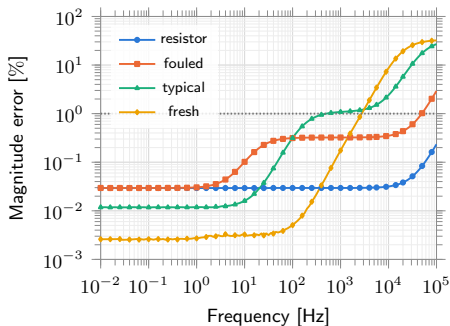
Vision

Against Moshkin and the field

From currents to a score

References

Backup



Accuracy $\pm 1\%$, $\pm 0.6^\circ$ to 5 kHz; $\pm 5\%$ to 27 kHz. Limit: A1's loop bandwidth
 Range 246 Ω to 12.1 M Ω . Limits: TIA compliance below, one LSB of output above; 320 R_f codes in between

*schematic sim

The excitation amplitude never enters: Z is the ratio of two records through one converter, so DAC gain and droop cancel.

What sets the EIS error, and what averaging buys

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

Against Muskin and the field

From currents to a trace

References

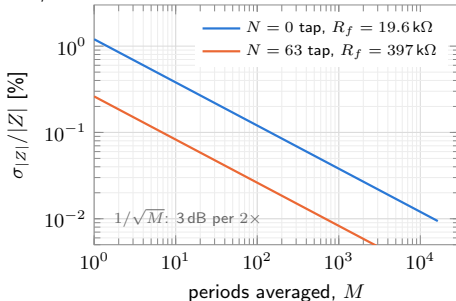
Backup

$$\varepsilon_{|Z|} \approx \underbrace{\frac{R_f/|Z|}{A_{\text{eff}}}}_{\text{systematic}} + \underbrace{\frac{i_n |Z|}{V_1} \sqrt{\frac{2}{NM}}}_{\text{random}}$$

$R_f/ Z $	feedback resistor over the load: the gain code chosen
A_{eff}	effective loop gain, ≈ 76 dB measured (0.66 % at $R_f/R = 40$); $A_{\text{OL}} \approx 83$ dB
i_n	front-end current noise per sample: 273 nA at $N=0$, 61 nA at $N=63$
V_1	excitation fundamental, $\frac{4}{\pi} \frac{A}{2}$
N, M	samples per period, periods averaged

The systematic term is fixed for a gain code and load, and the Z_{cal} ratio removes it (next slide). The random term is what remains, and it falls as $1/\sqrt{M}$.

Random term against periods averaged, 11 k Ω load, 1 kHz



Calibrating impedance: what cancels, and what a 20 % poly costs

$$\hat{Z} = Z_{\text{cal}} \frac{(\hat{V}_P / \hat{V}_{\text{OUT}})}{(\hat{V}_P / \hat{V}_{\text{OUT}})_{\text{cal}}} \quad \text{same gain code, same frequency, same converter}$$

	On-chip Z_{cal} (per decade, muxed in)	Off-chip standard (at test)
Cancels	R_f , its pole, the DAC, the LSB, A2's gain floor	the same
Value known	to the poly's chip-to-chip spread, ~20 %	to the standard, < 0.1 %
When	any time, in the field, before every point	once per chip, stored in flash
Tracks	temperature and ageing, since Z_{cal} and R_f are the same poly	the channel on the day it was tested
Good for	trends: fouling is a change in one electrode's Z , untouched by a fixed scale	absolute ohms

The poly need not be accurate, only stable. One off-chip number per chip pins the absolute scale; the on-chip ratio carries it through temperature and time.

The class-AB amplifier: one cell, both roles

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

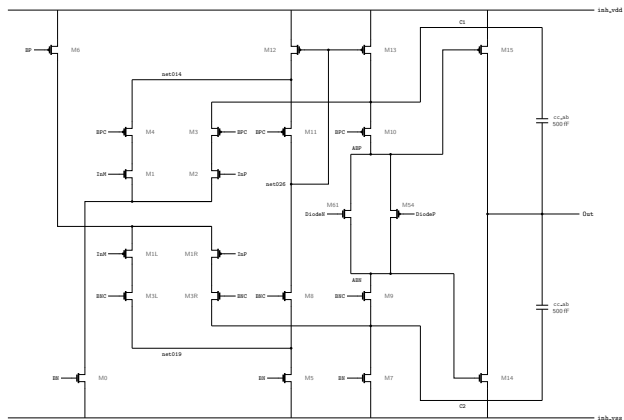
Vision

Against Mvshkin and the field

From currents to a score

References

Backup



complementary
input pairs and
their cascodes

diode-connected
summing branch

class-AB
drive

output

testbench
compensation

Bipolar accuracy through zero

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

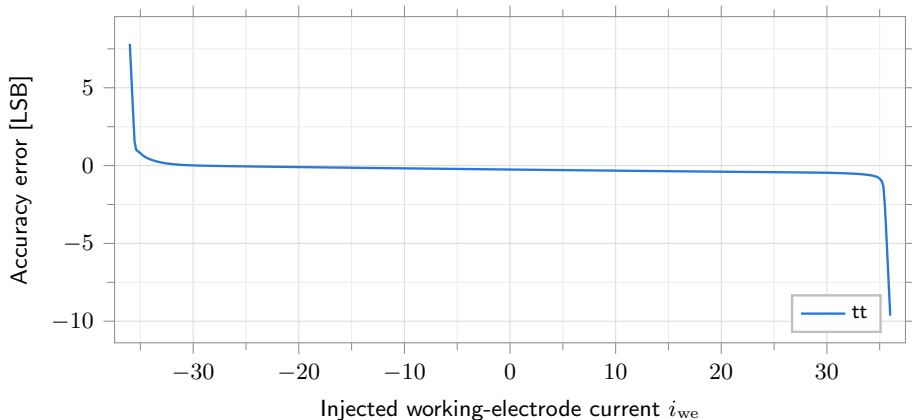
Vision

Against Moshkin and the field

From currents to a score

References

Backup



*schematic sim

No kink at the crossing: class-AB output, no dead zone. This is the measurement rev-1's PMOS mirror could not make.

Programmable feedback resistor: one string, 320 codes

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

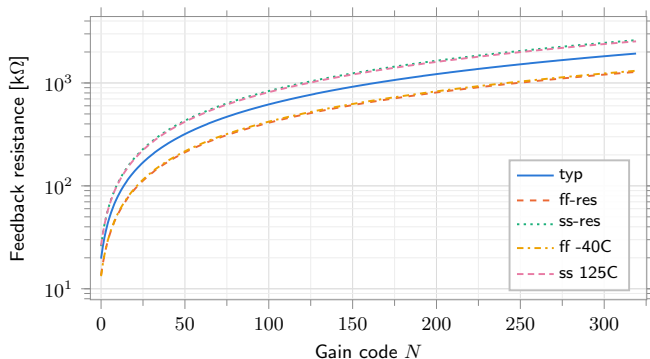
Vision

Against Muskhin and the field

From currents to a score

References

Backup



$$R_f = 18 \text{ k}\Omega + 6 \text{ k}\Omega \cdot N$$

N	R_f	Full scale
0	19.6 kΩ	±40.7 μA
63	397 kΩ	±2.02 μA
319	1.93 MΩ	±0.41 μA
LSB at 319		0.824 nA

Strictly monotonic at every corner, including the four thermometer seams. Maximum gain is noise-limited, not quantisation-limited.

10-bit SAR ADC, extracted netlist

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

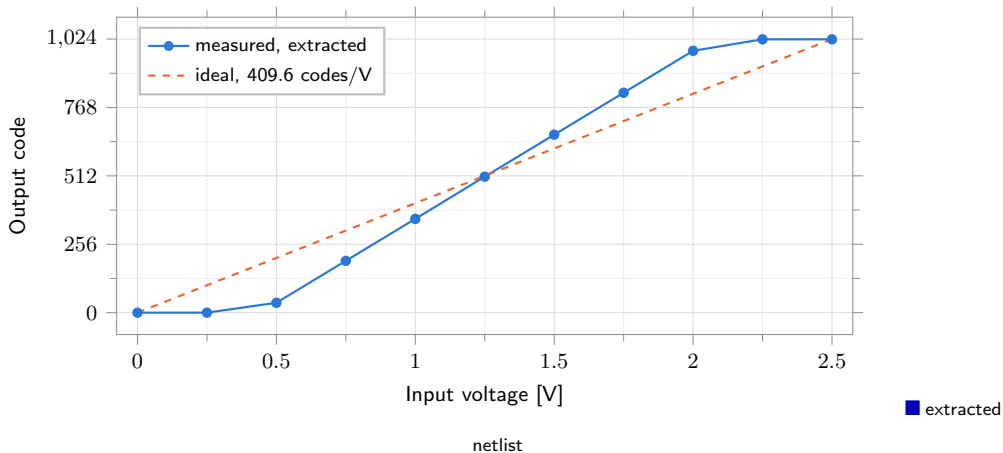
Vision

Against Moshkin and the field

From currents to a score

References

Backup



23 clocks per conversion, 870 kSa/s, 50 pJ. The capacitor array exists only in layout, so this is the one block characterised extracted.

Block results: corners, with mismatch σ

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

Against Munkin and the field

From currents to a score

References

Backup

Parameter	Unit	tt	ff	fs	sf	ss	Spread	σ (MC)
Class-AB amplifier (A1 and A2)								
Open-loop gain	dB	82.77	79.88	83.14	83.37	84.94	5.06	1.06
Phase margin	deg	73.52	74.70	74.04	72.86	73.34	1.84	1.42
Dominant pole	kHz	1.592	2.286	1.538	1.494	1.225	1.061	0.276
Gain-bandwidth product	MHz	22.78	23.19	22.89	22.94	22.59	0.60	3.57
Input offset voltage	mV	0.622	0.649	0.624	0.569	0.456	0.193	5.201
Programmable feedback resistor^a								
R at $N=0$	k Ω	19.64	—	—	—	—	—	0.071
R at $N=319$	M Ω	1.931	—	—	—	—	—	0.0007
Seam step 63 $\rightarrow$ 64	k Ω	7.42	—	—	—	—	—	0.504
10-bit SAR ADC, extracted^b								
Fitted slope	codes/V	628.59	628.59	628.40	628.73	628.13	0.60	—
Effective LSB	mV	1.5909	1.5909	1.5914	1.5905	1.5920	0.0015	—
Input at code 0	V	0.4408	0.4408	0.4407	0.4408	0.4399	0.0009	—
Usable span	V	1.6274	1.6274	1.6280	1.6271	1.6286	0.0015	—
Worst residual	LSB	0.495	0.495	0.490	0.477	0.484	0.018	—
RMS residual	LSB	0.294	0.294	0.277	0.288	0.302	0.025	—
Analog supply current	μ A	6.36	4.79	5.45	8.44	15.43	10.64	—
Energy per conversion	pJ	20.8	19.3	18.1	27.3	49.5	31.4	—
12-bit R-2R reference DAC, ideal rail								
Full-scale output	V	2.499388	2.499388	2.499388	2.499388	2.499388	0	0.000001
LSB size	μ V	610.351	610.351	610.351	610.351	610.351	0	0.0003
INL, most positive	LSB	0.113	0.101	0.100	0.127	0.130	0.030	0.610
INL, most negative	LSB	-0.376	-0.338	-0.390	-0.362	-0.414	0.076	0.620
DNL, most positive	LSB	0.002	0.001	0.002	0.002	0.002	0.000	1.128
DNL, most negative	LSB	-0.423	-0.374	-0.424	-0.424	-0.479	0.105	1.415

σ : 200 samples, process + mismatch, nominal corner. Corner runs hold the passive models at typical, so resistor and ladder spreads are lower bounds. ^aThe TRM carries R_f corners as a figure (the R_f slide), not a table. ^bExtracted netlist; the extraction carries no mismatch statistics, so no σ . Amplifier and DAC: schematic sim*; SAR: extracted.

One AFE channel, as simulated

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

Against Muskin and the field

From currents to a scan

References

Backup

Parameter	Value	Condition
CE compliance	13.2 mV – 2.480 V	2.5 V rail, corners
Cell current delivered	$\pm 52.5 \mu\text{A}$	12 k Ω cell
TIA compliance	$\pm 0.80 \text{ V}$ about v_{cm}	
Feedback resistor	19.6 k Ω – 1.93 M Ω , 320 codes	monotonic at every corner
Full-scale current	$\pm 40.7 \mu\text{A}$ ($N=0$) to $\pm 0.41 \mu\text{A}$ ($N=319$)	SAR window $\cap$ TIA compliance
Current quantum	0.824 nA at maximum gain	below the 2 nA noise budget
Bipolar accuracy	0.25–0.38 LSB worst	560 k Ω , 1 M Ω electrode, corners
Control-loop PM	79–80°	five corners, into a cell
TIA-loop PM	89–93°	12-cell electrode grid
Input current noise	341 pA standalone, 3.6–21 nA into an electrode	0.1–100 Hz, 560 k Ω
Zero-current offset	$\sigma(v_{\text{re}}) = 5.72 \text{ mV}$, gain-independent	200 MC samples, calibratable
Quiescent current	197–250 μA per channel	amplifiers + local bias

*schematic sim ■ Monte Carlo, 200 samples

Pre-layout, electrode held nominal: no row includes electrode-to-electrode spread.

Castalia at a glance

Castalia

Maxx Seminario

Digital

Harts	5 (1 orchestrator, 4 channel tiles)
ISA	rv32imac_zba_zbb_zbs_zbc
Clock	24 MHz rated, closed at 25 MHz
Measured CPI	1.260 aggregate, nine benchmarks
Private TCM	8 KiB per hart, 1-cycle
Shared RAM	80 KiB behind a round-robin arbiter
Synchronisation	16 hardware mutexes, LR/SC, AMOs
Interrupts	claim/complete, any vector to any hart
Accelerator	NPU: MLP, conv-1D, XNOR, GEMM
Power domains	4 MTCMOS, one per channel tile
Debug	JTAG, RISC-V Debug Module
Field boot	NFC ISO 14443A harvested boot

Analog and physical

Measurement sites	4 identical, simultaneous
Topology	bipolar grounded-WE potentiostat
Electrode pads	12 (CE/RE/WE $\times$ 4)
Converter	10-bit SAR per channel, extracted
Reference DAC	12-bit R-2R per channel
Feedback resistor	$18\text{ k}\Omega + 6\text{ k}\Omega \cdot N$, 320 codes
Full scale	$\pm 40.7\text{ }\mu\text{A}$ to $\pm 0.41\text{ }\mu\text{A}$
EIS	firmware mode of the same channels
Analog supply	2.5 V, $v_{\text{cm}} = 1.25\text{ V}$
Process	65 nm CMOS, 1P8M, HVT cells
Die / package	$3.0 \times 3.0\text{ mm}$ / LQFP-100
Digital power	10.72 mW post-route, SS, 25 MHz

Digital numbers from the hardened flow; analog numbers schematic-level, except the SAR, which is extracted.

Digital, as closed

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

Against Mushkin and the field

From currents to a score

References

Backup

Whole chip, post-route

Internal	3.78 mW
Switching	2.33 mW
Leakage	4.61 mW

Total	10.72 mW
--------------	-----------------

One channel tile

Total	1.761 mW
Gated	0

Measured CPI, timed kernels

Benchmark	Instr.	CPI
spmv	802 447	1.143
multiply	20 892	1.163
vvadd	2 408	1.374
median	4 015	1.398
qsort	123 205	1.398
rsort	146 548	1.412
dhrystone	200 523	1.504
towers	3 747	1.682
memcpy	11 017	1.726
Aggregate	1 314 802	1.260

Timing at 25 MHz

reg2reg WNS	+99 ps
Leakage share	43 %

■ hardened flow, post-route

SS 0.9 V 125 °C, statistical activity 0.2. Gating one tile removes 1.76 mW; the always-on band pays nothing.

Power gating: one switched rail per channel tile

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

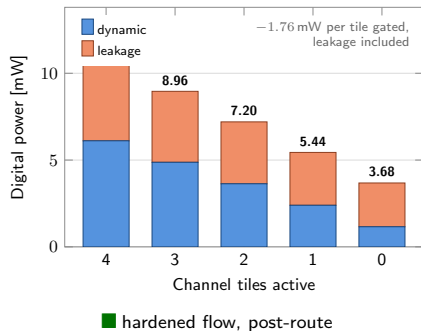
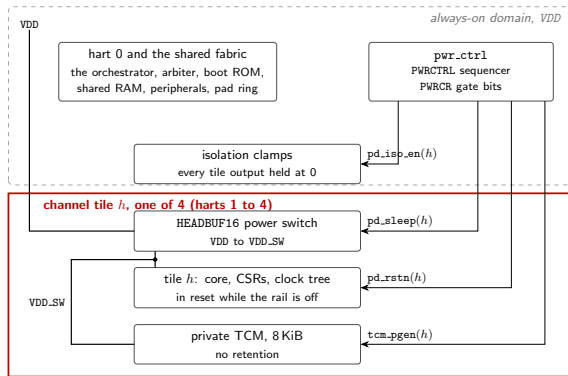
Vision

Against Musklin and the field

From currents to a score

References

Backup



Cold gating: the header cuts the rail, so a parked tile draws neither dynamic nor leakage power. Each tile gated removes 1.76 mW; the always-on band keeps 3.68 mW.

Vision: the same channel, without the cartridge

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

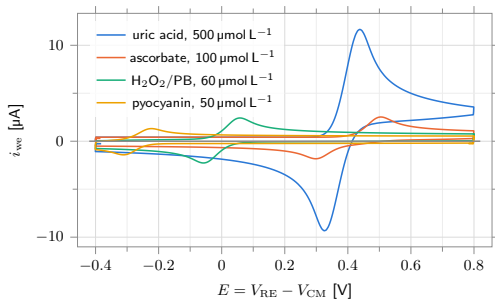
Vision

Against Moshkin and the field

From currents to a score

References

Backup



*schematic sim

Already on the die

4 sites	concurrent, each its own DAC, SAR, hart
EIS	electrode state without a second instrument
NFC	harvested boot, battery-free readout
MTCMOS	a parked site costs nothing
NPU	the decision stays on the patient

Left: a small-molecule wound panel, uric acid, ascorbate, H_2O_2 and pyocyanin, through the same channel.

The ovarian panel stays a serum test: CA-125 is an epitope on a 3–5 MDa mucin and does not reach interstitial fluid. The wearable is a different, reagentless panel on the same silicon; the implant is the same plus a fouling problem that EIS can watch but not solve.

Against the field: the ISCAS'26 table, plus Castalia

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

Against Myshkin and the field

From currents to a score

References

Backup

Parameter	Castalia (this work)	Myshkin [9]	Chen '19 [10]	Ghoreishizadeh '17 [11]	Lin '23 [12]
Modality	CA/CV/DPV/EIS	CA/CV	CA/CV	CA/CV	CV/EIS
Channels	4, simultaneous	1	1	1, differential	1
Current direction	bipolar	unipolar	bipolar	bipolar	bipolar
Sensitivity (pA)	341 ^a	56.5 ^b	41	0.47	1.7
$I_{\max}$ (μA)	±40.7	+20	±5	±20	-48/+16
Power (mW)	3.68 ^c	3.0 ^d	0.016	9.3	1.0
Process (nm)	65	65	180	350	180
On-chip processor	5 RISC-V harts	1 RISC-V hart	none	none	none
On-chip ML	NPU, 4 modes	NPU, 1 mode	none	none	none

Castalia column is best case and pre-silicon. ^aAFE only: input-referred rms, 0.1–100 Hz, maximum gain, ideal R_f , no electrode; the converter LSB is 0.82 nA. ^bAFE only; ADC quantisation limits the system to 153 pA. ^cDigital only: hart 0 awake, all four channel tiles gated, post-route SS 125 °C 25 MHz; one channel measuring adds 1.76 mW (tile) + 1.14 mW (AFE). ^dMeasured: 2.8 mW digital at 24 MHz + 258 μW AFE.

Still the only entries with a processor and an ML engine on the die; Castalia adds concurrency, both current polarities and EIS.

The fusion runs on the NPU

Castalia

Maxx Seminario

Target
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

Against Moshkin
and the field

From currents to
a score

References

Backup

$$s = \sigma \left(\sum_{k=1}^4 w_k \hat{c}_k + b \right)$$

Input	peak current, peak potential, baseline $\times 4$
Mode	0, dense layer, one sigmoid
Output	one risk score
Weights	trained off-chip, staged by firmware

Per site

Hart	1, ownership enforced in hardware
Potential	own 12-bit DAC
Converter	own 10-bit SAR
Power	own MTCMOS domain

The logistic form of the cleared two-marker algorithm, extended to four inputs and moved from the central laboratory onto the electrode.

References (1/5)

Castalia

Maxx Seminario

Target application

The chip

Analysis front end

Operating modes

Analysis blocks

Digital

Vision

Against Musklin and the field

From currents to a score

References

Backup

- [1] NCI, *SEER Cancer Stat Facts*, per-site pages, seer.cancer.gov/statfacts, accessed 20 Sep. 2026. Stage, survival: SEER 21, 2016–22; cases, deaths: 2026 est.
- [2] D. C. Grossman *et al.*, "Screening for ovarian cancer: USPSTF recommendation statement," *JAMA* 319:588, 2018.
- [3] USPSTF, *A and B Recommendations; Prostate Cancer: Screening*, 2018; *Pancreatic Cancer: Screening*, 2019. uspreventiveservicestaskforce.org.
- [4] U. Menon *et al.*, "Ovarian cancer population screening and mortality after long-term follow-up in UKCTOCS," *Lancet* 397:2182, 2021.
- [5] F. Chen *et al.*, "Clinical analysis of four serum tumor markers in 458 patients with ovarian tumors: combined use of HE4, CA125, CA19-9, and CEA," *Cancer Manag. Res.* 10:1313, 2018.
- [6] R. G. Moore *et al.*, "A novel multiple marker bioassay utilizing HE4 and CA125 for the prediction of ovarian cancer in patients with a pelvic mass," *Gynecol. Oncol.* 112:40, 2009.

References (2/5)

Castalia

Maxx Seminario

Target application

The chip

Analogue front end

Operating modes

Analogue blocks

Digital

Vision

Against Musklin and the field

From currents to a SoC

References

Backup

- [7] USPSTF, *Grade Definitions* (current, after July 2012), uspreventiveservicestaskforce.org, accessed 20 Sep. 2026.
- [8] P. Charkhchi *et al.*, "CA125 and ovarian cancer: a comprehensive review," *Cancers* 12:3730, 2020.
- [9] M. A. Seminario, S. McRobert, A. Uerling, P. Aberson, S. Balkır, J. A. Schmitz and E. J. Markvicka, "An electrochemical sensing SoC for autonomous wound monitoring," *IEEE ISCAS*, 2026.
- [10] Y.-C. Chen, S.-Y. Lu and Y.-T. Liao, "A microwatt dual-mode electrochemical sensing current readout with current-reducer ramp waveform generation," *IEEE TBioCAS* 13(6):1163, 2019.
- [11] S. S. Ghoreishizadeh *et al.*, "A differential electrochemical readout ASIC with heterogeneous integration of bio-nano sensors for amperometric sensing," *IEEE TBioCAS* 11(5):1148, 2017.
- [12] Y.-J. Lin *et al.*, "A multimodality electrochemical and impedance spectroscopy system-on-a-chip with temperature sensing and impedance-boosting techniques," *IEEE TBioCAS* 17(4):857, 2023.

References (3/5)

Castalia

Maxx Seminario

Target
application

The chip

Analogue front end

Operating modes

Analogue blocks

Digital

Vision

Against Muskin
and the field

From currents to
a score

References

Backup

- [13] A. J. Bard and L. R. Faulkner, *Electrochemical Methods: Fundamentals and Applications*, 2nd ed., Wiley, 2001 (Eqs. 3.3.11, 4.4.2, 4.4.29, Sec. 1.4.2, Eq. 6.2.19).
- [14] M. S. Wilson and W. Nie, "Multiplex measurement of seven tumor markers using an electrochemical protein chip," *Anal. Chem.*, 2006.
- [15] A. Kovářová *et al.*, "Biosensor with electrochemically active nanocomposites for signal amplification and simultaneous detection of three ovarian cancer biomarkers," *Electrochim. Acta*, 2023.
- [16] V. Mirčeski *et al.*, "Square-wave voltammetry: a review on the recent progress," *Electroanalysis*, 2013.
- [17] D. Britz and J. Strutwolf, *Digital Simulation in Electrochemistry*, 4th ed., Springer, 2016 (Ch. 3 box method, Ch. 7 expanding grid).
- [18] S. W. Feldberg, "Digital simulation: a general method for solving electrochemical diffusion-kinetic problems," *Electroanal. Chem.* 3:199, 1969; J. E. B. Randles, *Discuss. Faraday Soc.* 1:11, 1947.

References (4/5)

Castalia

Maxx Seminario

Target
application

The chip

Analysis front end

Operating modes

Analysis blocks

Digital

Vision

Against Muskin
and the field

From currents to
a scan

References

Backup

- [19] Jog *et al.*, Verilog-A three-electrode model with kinetics and mass transport for co-simulation with interface circuits, *IEEE SENSORS*, 2024, doi:10.1109/SENSORS60989.2024.10784491; S. Madec *et al.*, *IEEE TCAS-I* 66(6):2188, 2019.
- [20] P. Fanjul-Bolado, M. B. González-García and A. Costa-García, "Amperometric detection in TMB/HRP-based assays," *Anal. Bioanal. Chem.* 382:297, 2005.
- [21] M. Bilgi Kamaç *et al.*, "Point-of-care testing: a disposable label-free electrochemical CA125 and HE4 immunosensors for early detection of ovarian cancer," *Biomed. Microdevices* 25:18, 2023.
- [22] D. Kang, F. Ricci, R. J. White and K. W. Plaxco, "Survey of redox-active moieties for application in multiplexed electrochemical biosensors," *Anal. Chem.* 88(21):10452, 2016.
- [23] V. Clark, M. Aller Pellitero and N. Arroyo-Currás, "Explaining the decay of nucleic acid-based sensors under continuous voltammetric interrogation," *Anal. Chem.* 95(11):4974, 2023.
- [24] K. K. Leung *et al.*, "Elucidating the mechanisms underlying the signal drift of electrochemical aptamer-based sensors in whole blood," *ACS Sens.* 6:3340, 2021.

References (5/5)

Castalia

Maxx Seminario

Target application

The chip

Analogue front end

Operating modes

Analogue blocks

Digital

Vision

Against Muskin and the field

From currents to a topic

References

Backup

[25] H. Li *et al.*, "Dual-reporter drift correction to enhance the performance of electrochemical aptamer-based sensors in whole blood," *J. Am. Chem. Soc.* 138:15809, 2016.

[26] J. Sabatè del Río *et al.*, "An antifouling coating that enables affinity-based electrochemical biosensing in complex biological fluids," *Nat. Nanotechnol.* 14:1143, 2019.

[27] Manufacturer labelling and FDA clearances: Dexcom G7 (10-day; 15-day, adults), Abbott FreeStyle Libre 3 (14 days), Senseonics Eversense 365 (implanted, 365 days; FDA clearance 17 Sep. 2024).

[28] E. Shirzaei Sani *et al.*, "A stretchable wireless wearable bioelectronic system for multiplexed monitoring and combination treatment of infected chronic wounds," *Sci. Adv.* 9(12):eadf7388, 2023.

[29] Son *et al.*, "Continuous, week-long, seconds-resolved in vivo drug measurements performed with a xenonucleic acid-employing electrochemical, aptamer-based sensor," *J. Am. Chem. Soc.* 148(9):10016, 2026.

Backup: electrode model validation

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

Against Muskin
and the field

From currents to
a score

References

Backup

Scan rate mV s^{-1}	i_p Randles–Ševčík μA	i_p model μA	Error %	Error, $C_{\text{dl}} \rightarrow 0$ %	ΔE_p mV
25	0.951	0.952	+0.08	−0.45	58.78
50	1.345	1.349	+0.32	−0.42	59.85
100	1.902	1.915	+0.67	−0.38	60.41
200	2.690	2.721	+1.16	−0.33	62.33
400	3.804	3.875	+1.86	−0.25	64.74

Peak current against Randles–Ševčík and ΔE_p against Nicholson [13], reversible couple, $C_{\text{dl}} = 0$ column isolates the faradaic model.

Backup: USPSTF screening grades

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

Against Musklin and the field

From currents to a score

References

Backup

Grade	Certainty of net benefit	Definition [7]	Cancer screening [2, 3]
A	high, substantial	Recommends the service. Offer or provide it.	cervix (2018), colorectal 50–75 (2021)
B	high moderate, or moderate	Recommends the service. Offer or provide it.	breast 40–74 (2024), lung LDCT 50–80 (2021), colorectal 45–49 (2021) prostate PSA 55–69 (2018)
C	at least moderate, small	Offer selectively, by professional judgment and patient preference.	
D	no net benefit, or harms outweigh	Recommends <i>against</i> the service. Discourage its use.	ovary (2018) , pancreas (2019)
I	cannot be determined	Evidence insufficient, poor, or conflicting.	—
none	—	No USPSTF statement exists for the condition.	uterus

Ovarian D is a verdict on a test that was tried, CA-125 plus transvaginal ultrasound [2, 4], not on the absence of one.